Sample Plots for Each Neural Network

The following set of plots should be consistently generated for all basic neural network models, including ANN, LSTM, BNN, CVAE, and CGAN. These common plots ensure uniform evaluation and comparison across models.

Common Plots (Applicable to All Networks)

1. Density plots wrt inputs
2. Density plots wrt inputs
3. Histograms
4. Architecture plots
5. Prediction vs Actual (Regression plots)

   Scatter plot comparing predicted and true values, along with a 1:1 reference line.

6. Residual Analysis

   Residuals (predicted – actual) plotted against key parameters (e.g., period, magnitude, or distance).

7. Sensitivity analysis
8. SHAP and relative importance plots

Model-Specific Plots

In addition to the common plots, some figures will vary depending on the neural network architecture:

1. Network Architecture

   A schematic diagram illustrating the structure of each model (e.g., ANN, LSTM, etc.).

2. BERT-Based Model

   Include an additional plot showing the prediction of magnitude (Mw), rupture distance (Rrup), and Vs30 for given PSA values.

3. Additional Model-Specific Visualizations

   Include plots relevant to the specific characteristics of each model (e.g., uncertainty for BNN, latent space for CVAE, generated vs real data for CGAN).

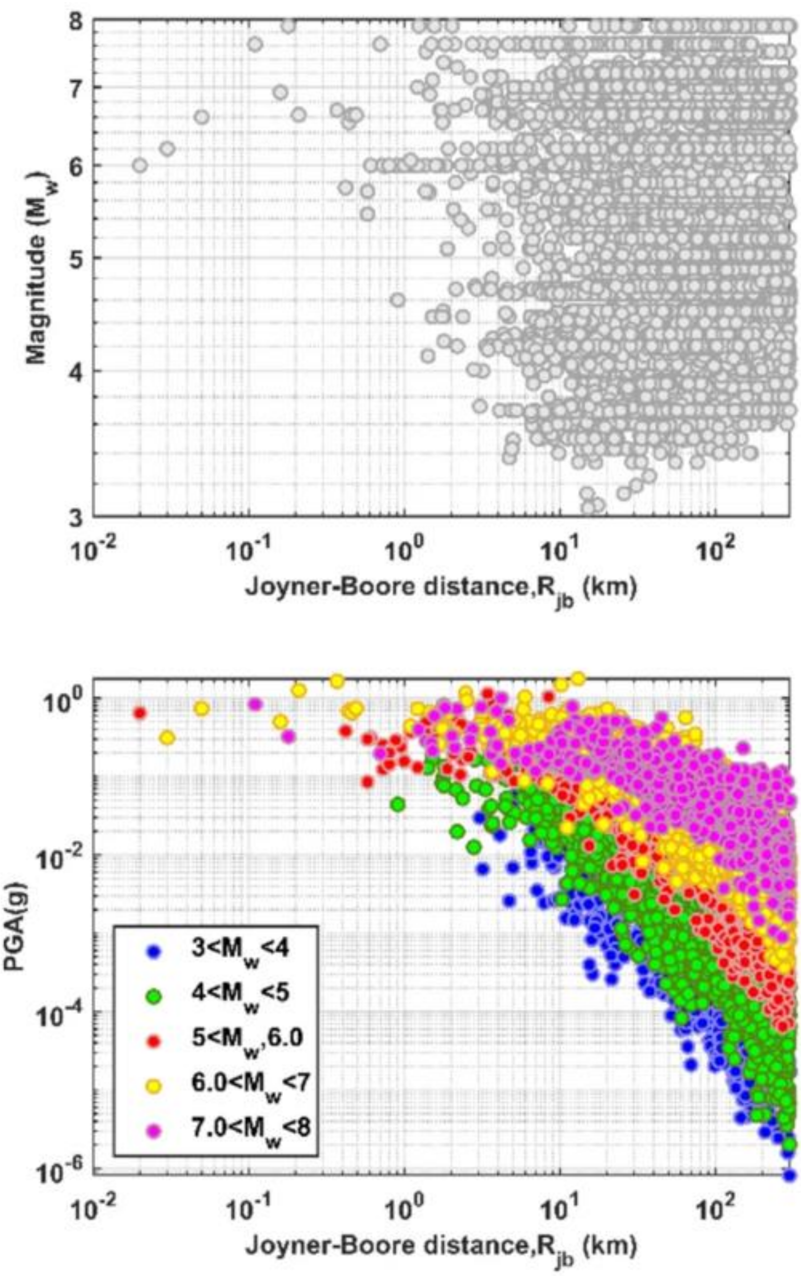

Fig 01: Data used(name)

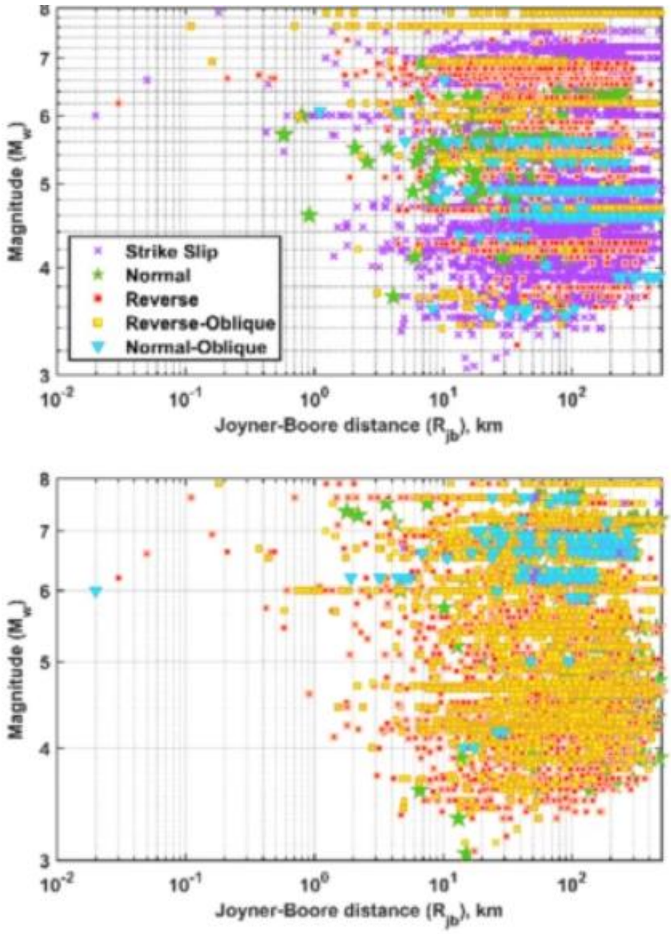

Fig 02: Data used (name of the data)

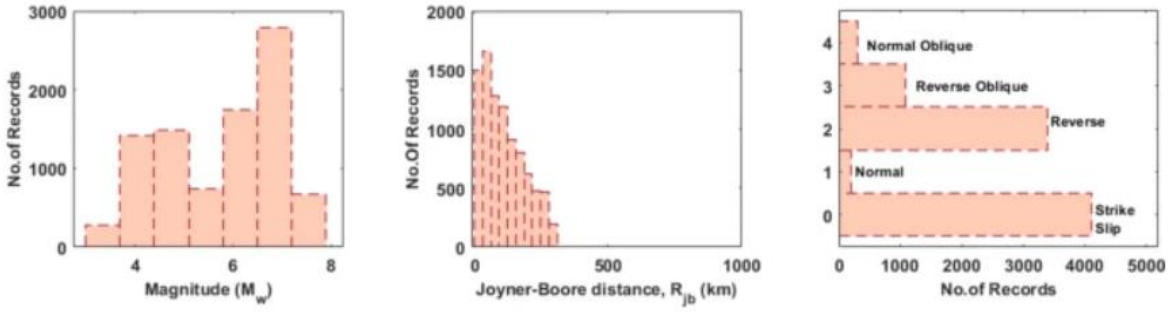

Fig 03: Frequency plots

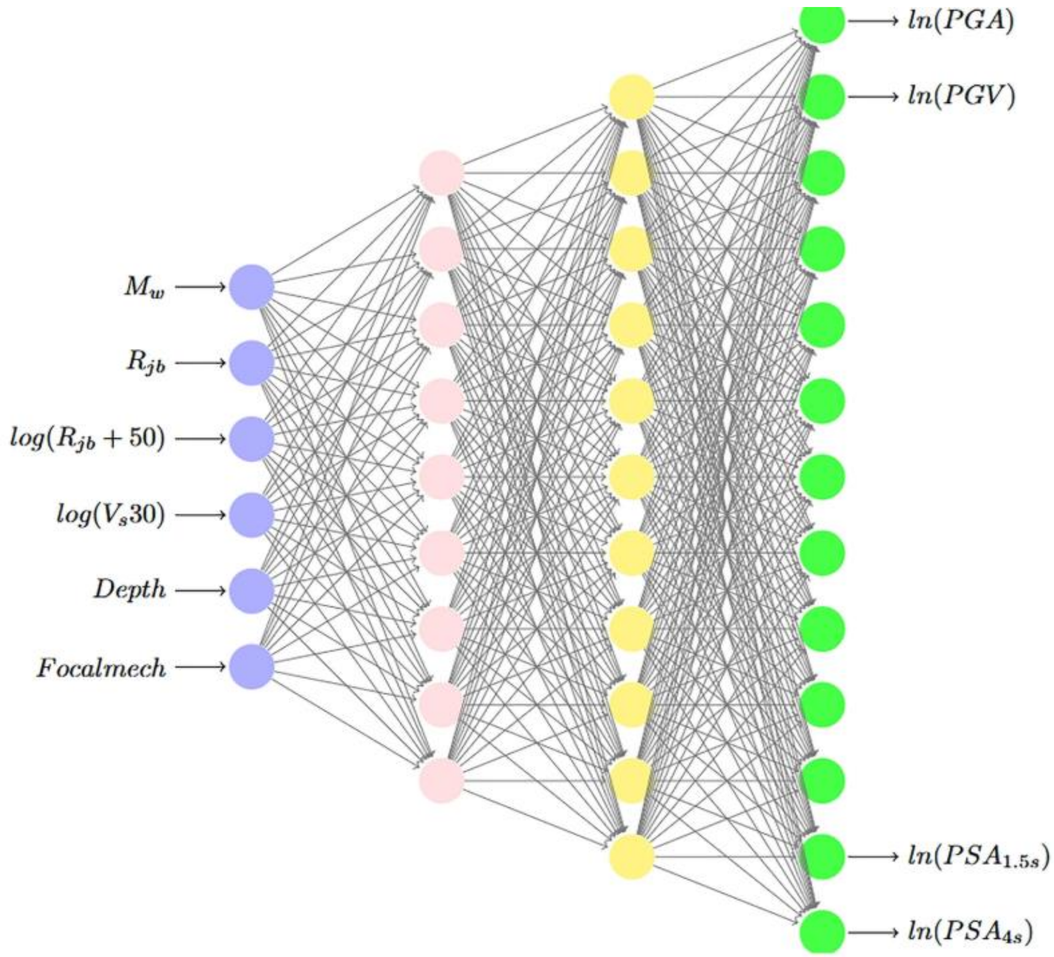

Fig 04: Architecture for ANN

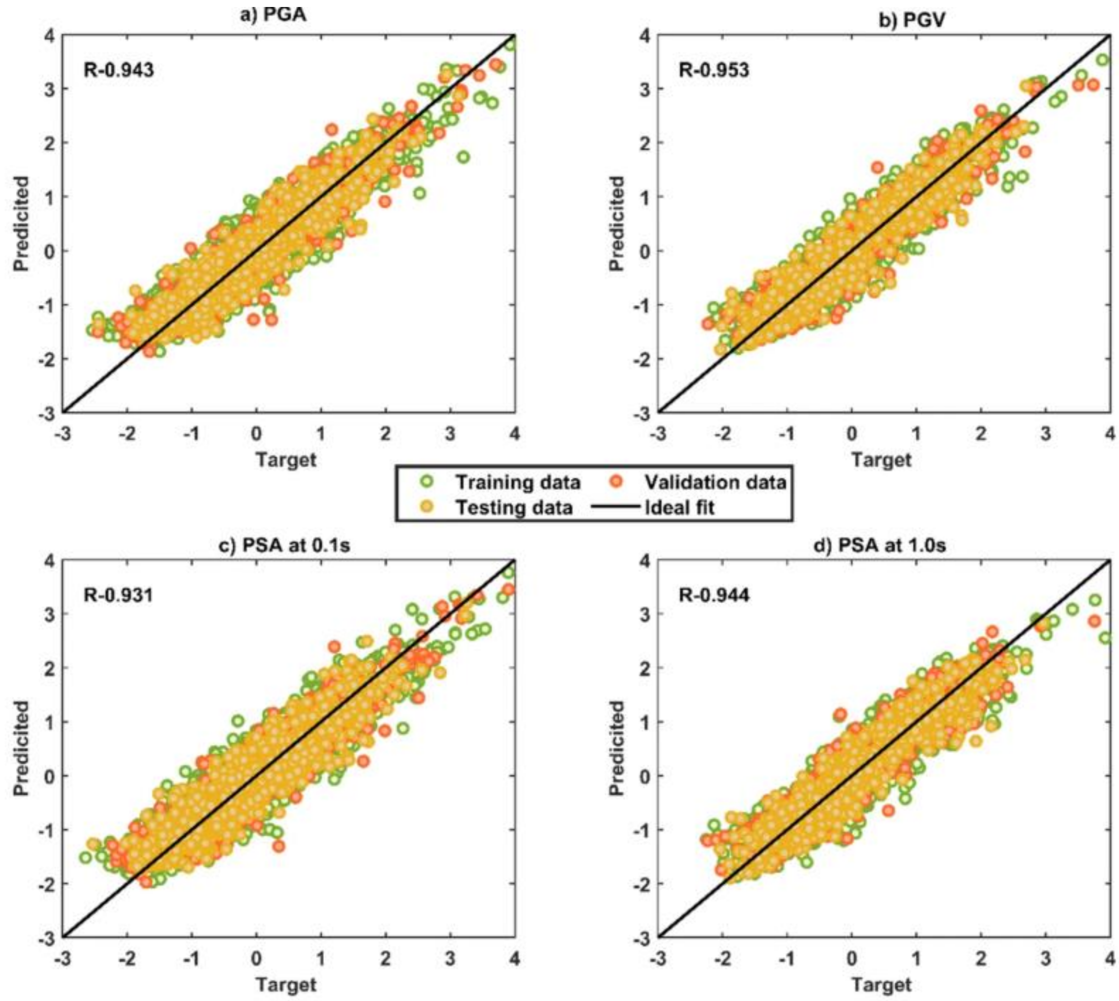

Fig 05: Regression plots

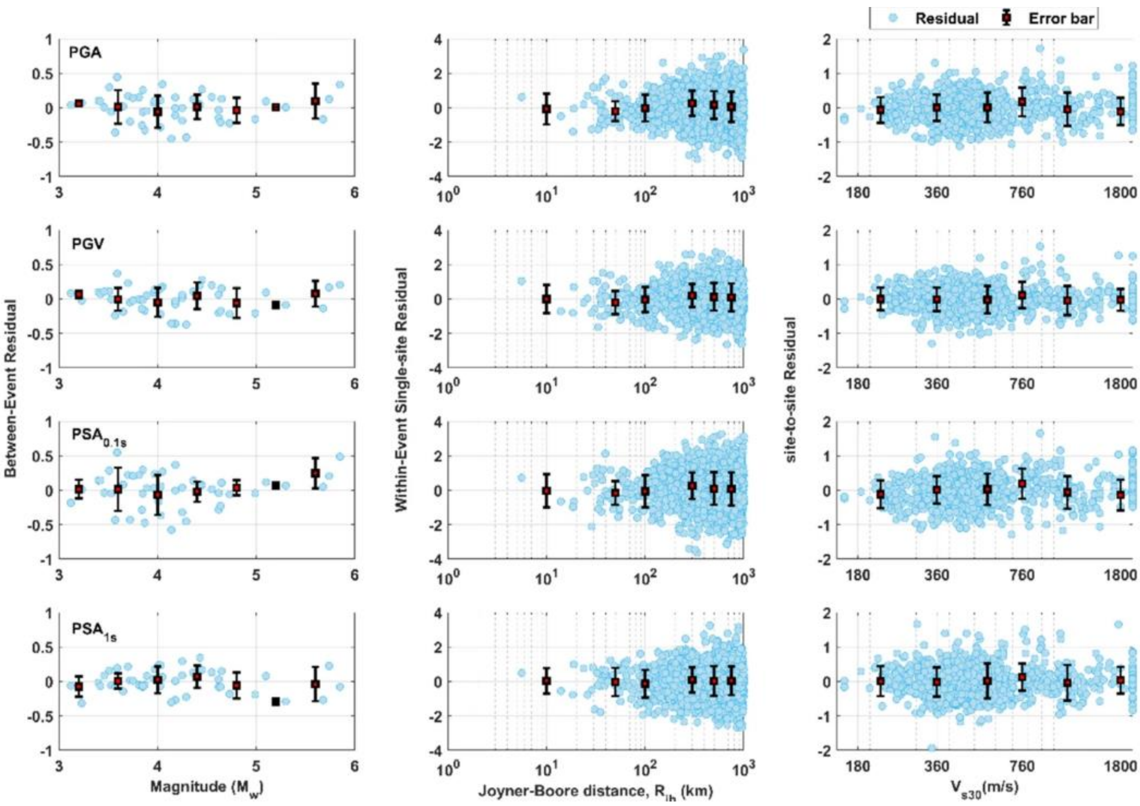

Fig 06: Residual plots

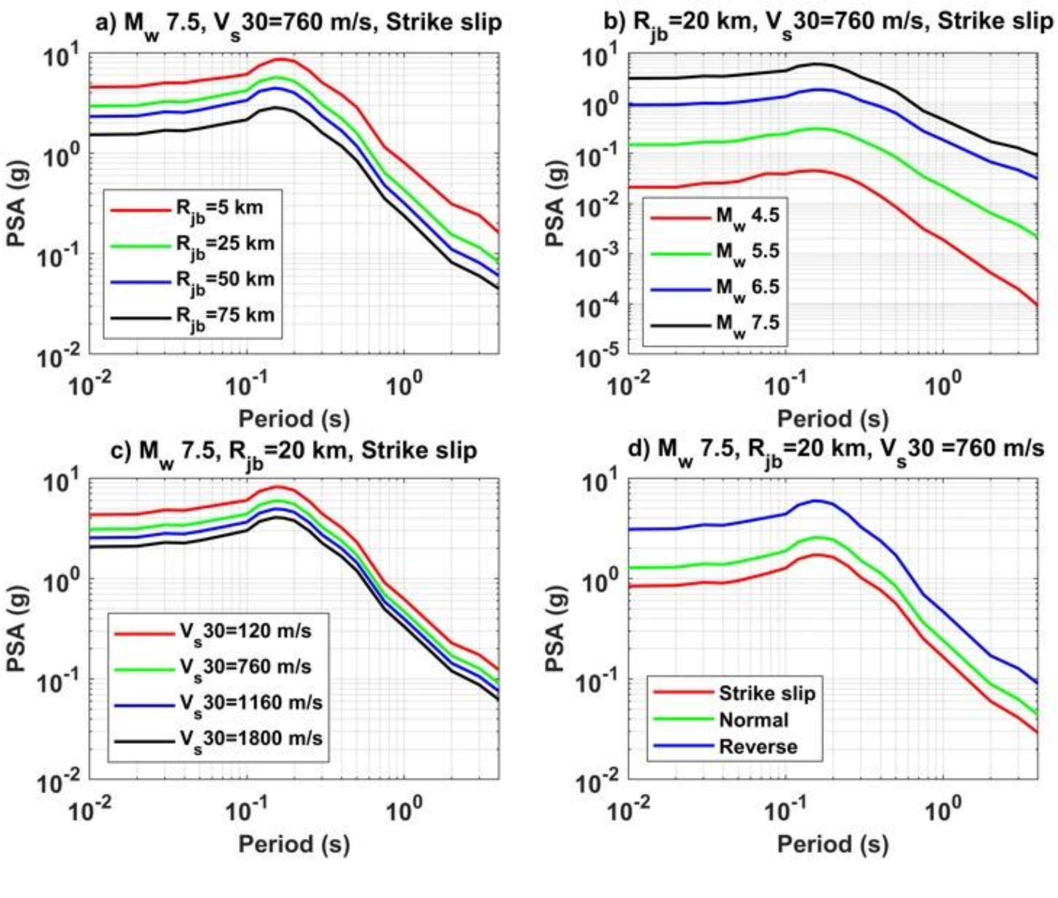

Fig 07: Sensitivity plots/ Physics plots

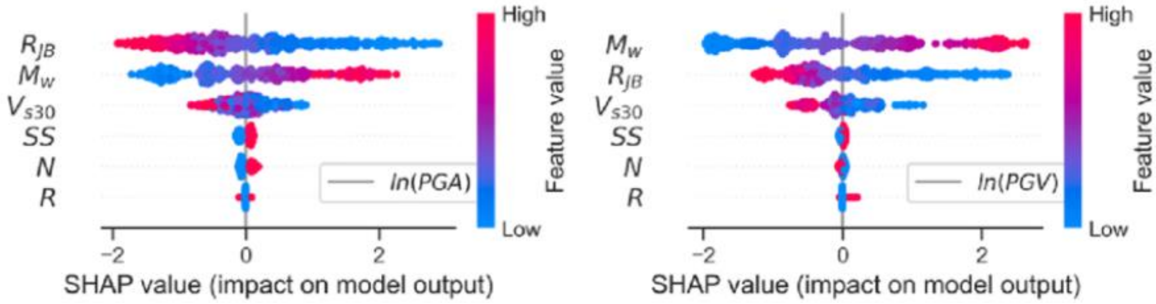

Fig 08: Shap analysis according to the inputs/outputs

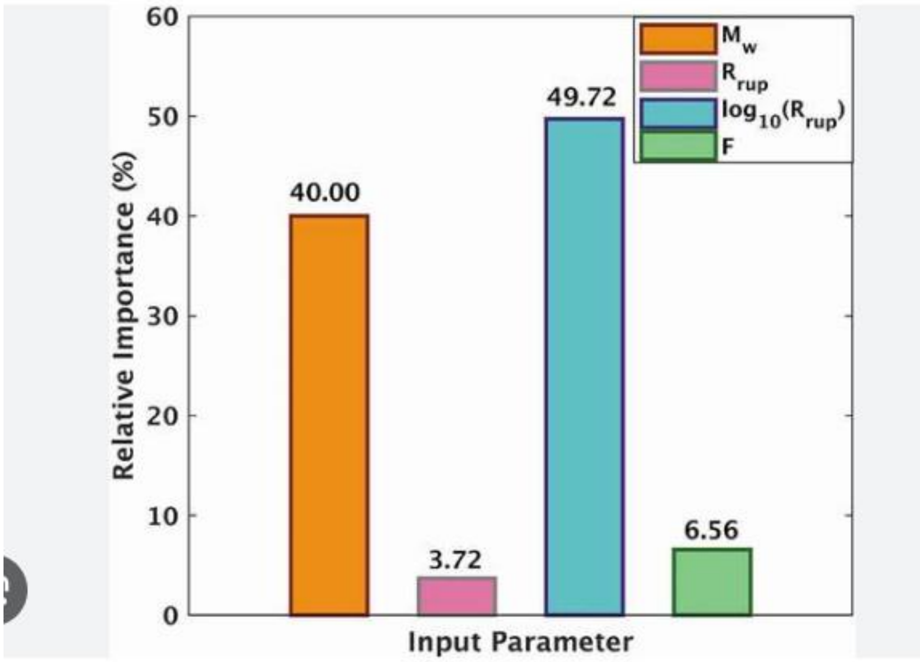

Fig 09: Relative importance

Table 01: Statistics of the data

Parameter	Min	Max	Mean	Median	STD	Skewness	Kurtosis
M_w	3.05	7.90	5.78	6.20	1.18	- 0.25	1.81
$-R_{jb}$	- 300.83	- 0.02	- 111.12	- 96.80	77.53	- 0.59	2.35
$-\log(R_{jb})$	- 5.71	3.91	- 4.35	- 4.57	1.01	1.38	6.36
$\log(V_{s30})$	4.49	7.65	5.99	5.96	0.40	0.12	3.27
Depth	0.02	22.00	10.22	9.56	3.82	0.41	2.62
PGA	- 14.02	0.57	- 4.71	- 4.34	2.27	- 0.69	3.16
PGV	- 8.72	4.80	- 0.23	0.29	2.45	- 0.64	2.77
$PSA_{0.01s}$	- 14.02	0.58	- 4.70	- 4.33	2.27	- 0.69	3.16
$PSA_{0.02s}$	- 14.02	0.61	- 4.70	- 4.32	2.27	- 0.69	3.16
$PSA_{0.03s}$	- 14.02	0.69	- 4.67	- 4.30	2.28	- 0.68	3.16
$PSA_{0.04s}$	- 14.02	0.85	- 4.62	- 4.26	2.29	- 0.67	3.16
$PSA_{0.05s}$	- 14.02	0.93	- 4.56	- 4.21	2.30	- 0.66	3.16
$PSA_{0.075s}$	- 14.01	1.07	- 4.36	- 4.03	2.32	- 0.67	3.20
$PSA_{0.10s}$	- 14.00	1.04	- 4.20	- 3.86	2.33	- 0.67	3.23
$PSA_{0.12s}$	- 13.99	1.05	- 4.10	- 3.78	2.32	- 0.67	3.24
$PSA_{0.15s}$	- 13.95	1.20	- 4.00	- 3.66	2.30	- 0.67	3.24
$PSA_{0.20s}$	- 13.97	1.21	- 3.97	- 3.63	2.29	- 0.67	3.22
$PSA_{0.25s}$	- 13.86	1.46	- 3.94	- 3.58	2.28	- 0.67	3.21
$PSA_{0.30s}$	- 13.58	1.44	- 3.94	- 3.59	2.27	- 0.66	3.15
$PSA_{0.40s}$	- 13.63	1.57	- 3.97	- 3.61	2.27	- 0.64	3.08
$PSA_{0.50s}$	- 13.29	1.21	- 4.10	- 3.73	2.28	- 0.61	2.99
$PSA_{0.60s}$	- 13.37	1.19	- 4.25	- 3.88	2.31	- 0.58	2.88
$PSA_{0.75s}$	- 12.99	1.22	- 4.61	- 4.21	2.39	- 0.54	2.73
$PSA_{0.85s}$	- 13.14	0.74	- 4.94	- 4.51	2.47	- 0.53	2.64
PSA_{1s}	- 14.25	0.04	- 5.92	- 5.33	2.71	- 0.53	2.43
$PSA_{1.5s}$	- 15.17	- 0.81	- 6.58	- 5.91	2.85	- 0.50	2.35
PSA_{4s}	- 15.89	- 1.21	- 7.06	- 6.32	2.95	- 0.49	2.29

Table 02: Residuals/standard deviations

Parameter	Between event δB	Site to site δS	Within event single event δWS	Total sigma σ	Single station sigma σ_{ss}
PGA	0.2048	0.3935	0.8022	0.9167	0.8279
PGV	0.1746	0.3481	0.7034	0.804	0.7247
$PSA_{0.01s}$	0.2049	0.3935	0.8021	0.9166	0.8279
$PSA_{0.02s}$	0.2078	0.3962	0.8049	0.9208	0.8313
$PSA_{0.03s}$	0.2172	0.3837	0.8177	0.929	0.8461
$PSA_{0.04s}$	0.2283	0.3903	0.8346	0.9492	0.8653
$PSA_{0.05s}$	0.2439	0.396	0.8519	0.9706	0.8861
$PSA_{0.075s}$	0.2517	0.3845	0.8627	0.9775	0.8987
$PSA_{0.10s}$	0.2521	0.4177	0.8848	1.0104	0.92
$PSA_{0.12s}$	0.2472	0.4445	0.8909	1.0259	0.9246
$PSA_{0.15s}$	0.2393	0.4654	0.8882	1.0309	0.9199
$PSA_{0.20s}$	0.254	0.5027	0.9101	1.0703	0.9448
$PSA_{0.25s}$	0.2683	0.5037	0.9127	1.0765	0.9513
$PSA_{0.30s}$	0.2616	0.4885	0.8965	1.0539	0.9339
$PSA_{0.40s}$	0.2476	0.4754	0.8783	1.0289	0.9125
$PSA_{0.50s}$	0.2198	0.4745	0.86	1.0066	0.8877
$PSA_{0.60s}$	0.2015	0.4601	0.8253	0.9661	0.8496
$PSA_{0.75s}$	0.1823	0.4513	0.8073	0.9427	0.8276
$PSA_{0.85s}$	0.1832	0.4353	0.801	0.9298	0.8216
PSA_{1s}	0.1647	0.4308	0.8018	0.9249	0.8185
$PSA_{1.5s}$	0.1645	0.3974	0.7575	0.8711	0.7752
PSA_{4s}	0.2375	0.3159	0.7202	0.8215	0.7583

Table 03: Performance parameters

Parameter	R	K	K'	R_0^2	$R_0'^2$
PGA	0.943	1.000	0.995	0.880	0.889
PGV	0.954	0.996	0.993	0.903	0.910
$PSA_{0.01s}$	0.943	1.000	0.995	0.880	0.889
$PSA_{0.02s}$	0.943	1.000	0.995	0.880	0.889
$PSA_{0.03s}$	0.941	1.000	0.995	0.874	0.885
$PSA_{0.04s}$	0.939	1.001	0.994	0.871	0.881
$PSA_{0.05s}$	0.938	1.001	0.994	0.867	0.879
$PSA_{0.075s}$	0.933	0.999	0.995	0.857	0.871
$PSA_{0.10s}$	0.931	0.998	0.996	0.851	0.867
$PSA_{0.12s}$	0.933	0.999	0.994	0.856	0.870
$PSA_{0.15s}$	0.934	1.000	0.994	0.862	0.872
$PSA_{0.20s}$	0.932	1.002	0.992	0.857	0.868
$PSA_{0.25s}$	0.930	1.003	0.990	0.852	0.865
$PSA_{0.30s}$	0.929	1.003	0.990	0.848	0.863
$PSA_{0.40s}$	0.930	1.000	0.994	0.847	0.865
$PSA_{0.50s}$	0.930	0.998	0.996	0.846	0.865
$PSA_{0.60s}$	0.933	0.996	0.998	0.853	0.871
$PSA_{0.75s}$	0.939	0.995	1.001	0.867	0.881
$PSA_{0.85s}$	0.942	0.995	1.001	0.877	0.887
PSA_{1s}	0.945	0.996	1.000	0.883	0.892
$PSA_{1.5s}$	0.949	0.997	0.999	0.891	0.901
PSA_{4s}	0.957	0.996	1.001	0.900	0.914